Coding Project 2: Credit Analytics

A short risk primer and project overview

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Part A What do we mean by risk?

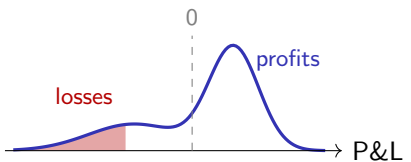
In everyday language, risk $\approx$ uncertainty. Risk management is more precise:

Risk = exposure to the possibility of loss

- **Exposure:** we hold a position, here a portfolio of loans.
- **Possibility:** its future P&L is random; we describe it with probabilities.
- **Loss:** what matters is the bad outcomes, not every surprise.

In Project 2

Borrower i repays with probability $p(x_i)$. Each loan is either repaid with interest or fully delinquent, so the portfolio P&L is random. Risk is the chance of (large) losses: the **left tail**.



Credit risk: losses from borrowers who **default**.

Q3: you will build this P&L distribution for three lending strategies.

Three ways to summarize risk (95% level)

Variance

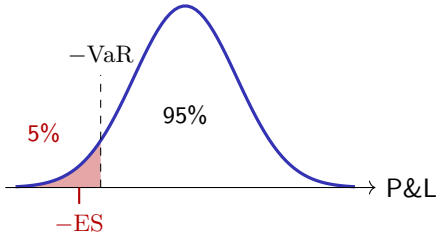
Average squared distance of the P&L from its mean: how **spread out** the outcomes are, counting gains and losses alike.

95%-Value-at-Risk (VaR)

$\mathrm{VaR}_{95\%} = -$ (5%-quantile of the P&L).
Only the worst 5% of scenarios lose more than VaR.

95%-Expected Shortfall (ES)

$\mathrm{ES}_{95\%} = -$ (average P&L over the worst 5% of scenarios).
If a bad 5% scenario happens, how much do we lose *on average*?



Project 2's convention: the minus sign turns a loss into a positive VaR. A **negative VaR means a gain**: even the 5%-worst scenario makes a profit.

Why VaR and ES rather than variance?

Toy example: two positions

	A	**B**
P&L outcomes	$+6$ / -6	$+2$ / -3 / -28
probabilities	50% / 50%	80% / 16% / 4%
mean	0	0
variance	36	36
95%-VaR	6	3
95%-ES	6	23

- **Variance**: identical, so it cannot tell A and B apart. It also counts gains as "risk".
- **VaR**: B looks *safer*. Its rare loss of 28 hides beyond the 5% threshold.
- **ES**: B is clearly *riskier*. ES looks into the tail.

In practice

VaR is simple and widely used (e.g. risk limits, regulation); ES is increasingly preferred because it sees the tail. Project 2 asks for **VaR**; ES is a natural extension.

Scenarios: Monte Carlo and stress testing

Monte Carlo scenarios (Project 2, Q3)

Simulate many "who repays" outcomes from a probability model. Each scenario gives one P&L; 50,000 scenarios give a distribution, and from it expected P&L and VaR (or ES).

Answers: how bad, and how often, *if the model is right*.

Stress scenario ("what if?")

Choose one specific adverse situation, e.g. a recession that pushes default rates up, and compute the P&L in that situation.

Answers: what happens in this state, without saying how likely it is.

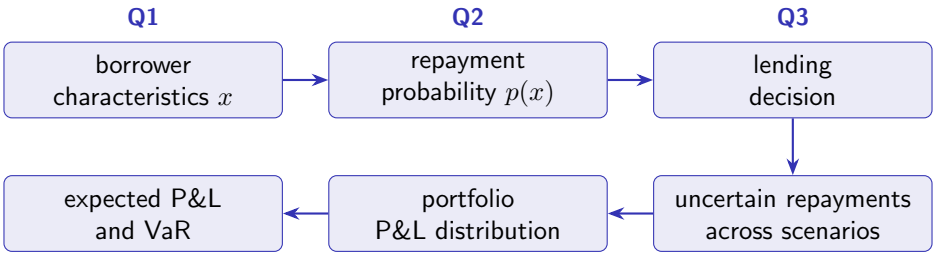
Why they complement each other

VaR and ES compress the model's own scenarios into one number. They are only as good as that model, and they do not show which situation causes the loss. Stress scenarios fill this gap, but they come without probabilities. Risk managers use both.

In Project 2: Q3 asks for the Monte Carlo view. Thinking in "what if" terms helps with the *Robustness* and *Limitations* parts of your report.

Part B Project 2 in one picture

Consumer lending: estimate who **repays**, turn predicted probabilities into **lending decisions**, and judge them by **P&L and risk**, not only by predictive accuracy.



Q1 Simulate the borrowers.
Q2 Predict repayment probabilities with two classifiers and compare them.
Q3 Lend with three strategies and measure the resulting P&L and risk.

Q1 and Q2 in a nutshell

Data

- Simulated borrowers: age, monthly income, self-employed. 20,000 for training, 10,000 for testing.
- Each borrower repays ($y = 1$) or defaults ($y = 0$) with a known probability $p(x)$.
- Two data sets, $s = 1, 2$: in p_1 the log-odds are linear in the features; in p_2 age acts through a "too young or too old" indicator.

Models

- **Logistic regression**: gives a repayment probability directly.
- **SVM** with a Gaussian kernel: gives a score; Platt scaling turns it into a probability.
- Compare the predicted probabilities: cross-entropy (train and test) and FDR/TPR curves.

Keep in mind

Both models are fitted to both data sets. Q2 asks how well each one fits, and why.

2c: what FDR and TPR measure

Decision rule: fund applicant i if $\hat{p}(x_i) \geq c$ ("positive" = predicted good, i.e. funded).

	good $(y = 1)$	**bad** $(y = 0)$
funded	TP	FP
not funded	FN	TN

- TPR $= \text{TP}/(\text{TP} + \text{FN})$: share of good borrowers who **get a loan**.
- FDR $= \text{FP}/(\text{TP} + \text{FP})$: share of funded loans that **turn out bad**.
- FPR $= \text{FP}/(\text{FP} + \text{TN})$: share of bad borrowers who get a loan (ROC curve).

Why FDR and not FPR (ROC)?

Losses come from bad loans *inside the lender's portfolio*: that share is FDR. FPR only counts how many of the (few) bad applicants slip through. ROC is a different curve, with a different AUC.

Building the curves (test data)

- One point (TPR, FDR) per threshold c; **TPR horizontal**.
- AUC of each plotted curve; points ordered by TPR.
- Nobody funded $\Rightarrow$ FDR $= 0/0$: state your convention.

Part of the question: explain how you construct the curves and what the AUC means in this setting.

Q3: three lending strategies

Applicants: the **test set**. Each loan is CHF 1000 and is either repaid with interest or fully delinquent.

Strategy	**Who gets a loan?**	**Interest**
(i)	everyone	5.5%
(ii)	logistic regression predicts repayment $\geq 95\%$	1%
(iii)	SVM predicts repayment $\geq 95\%$	1%

Report per strategy: P&L histogram, expected P&L, 95%-VaR (in CHF). Then compare.

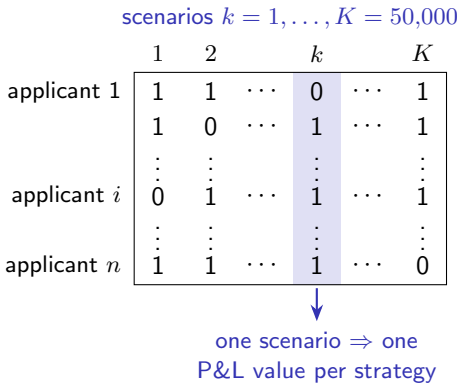
Two kinds of probabilities

The **model** decides who gets a loan; the **true** $p_2(x)$ decides who repays in the simulated scenarios.

Why train models if we know p_2?

p_2 plays the role of **reality**. A real lender never sees it and must learn $\hat{p}$ from past data. Knowing p_2 lets us simulate what lending with each model earns (expected P&L) and how risky it is (VaR).

The scenario matrix D



$$D_{i,k} = \begin{cases} 1 & \text{if } \xi_{i,k} \le p_2(x_{m+i}), \\ 0 & \text{otherwise.} \end{cases}$$

- **Row** i: one test applicant across all scenarios.
- **Column** k: one joint realization of all repayments ($1 =$ paid back with interest).
- A strategy's P&L in scenario k counts only the applicants it lends to. All strategies share the same D.

Practical

$n \times K = 5 \cdot 10^8$ entries (≈ 4 GB as float64): think about data types and blocks.

Schedule and further courses

Schedule

First discussion	Oct 14
Last questions	Oct 28
Deadline	Nov 4
In charge	Zhexin Wu

More risk management at ETH (RiskLab, D-MATH)

401-3631-00L Applied Risk Management

401-3629-00L Quantitative Risk Management